Chapter 2:
Operating-System Structures

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- Operating System Services
- User Operating System Interface
- System Calls
- Types of System Calls
- System Programs
- Operating System Design and Implementation
- Operating System Structure
- Virtual Machines
- Operating System Debugging
- Operating System Generation
- System Boot

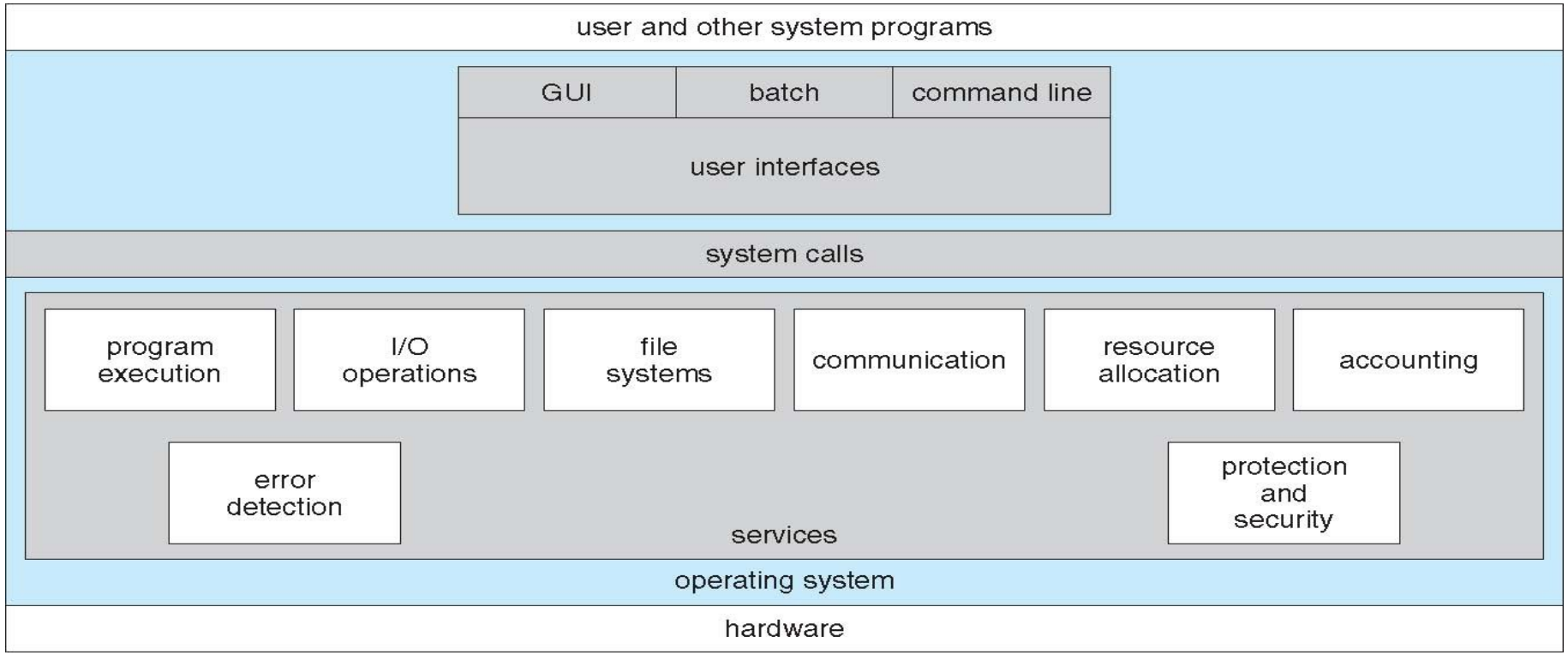
Objectives

- To describe the services an operating system provides to users, processes, and other systems
- To discuss the various ways of structuring an operating system
- To explain how operating systems are installed and customized and how they boot

Operating System Services

- One set of operating-system services provides functions that are helpful to the user:
  - **User interface** - Almost all operating systems have a user interface (UI)
    - Varies between **Command-Line (CLI)**, **Graphics User Interface (GUI)**, **Batch Interface.**
  - **Program execution** - The system must be able to load a program into memory and to run that program, end execution, either normally or abnormally (indicating error)
  - **I/O operations** - A running program may require I/O, which may involve a file or an I/O device
  - **File-system manipulation** - The file system is of particular interest. Obviously, programs need to read and write files and directories, create and delete them, search them, list file Information, permission management.

A View of Operating System Services



Operating System Services (Cont.)

- One set of operating-system services provides functions that are helpful to the user (cont.):
  - Communications – Processes may exchange information, on the same computer or between computers over a network
    - Communications may be via shared memory or through message passing (packets moved by the OS)
  - Error detection – OS needs to be constantly aware of possible errors
    - May occur in the CPU and memory hardware, in I/O devices, in user program
    - For each type of error, OS should take the appropriate action to ensure correct and consistent computing
    - Debugging facilities can greatly enhance the user's and programmer's abilities to efficiently use the system

Operating System Services (Cont.)

- **Resource allocation** - When multiple users or multiple jobs running concurrently, resources must be allocated to each of them
  - Many types of resources - Some (such as CPU cycles, main memory, and file storage) may have special allocation code, others (such as I/O devices) may have general request and release code
- **Accounting** - To keep track of which users use how much and what kinds of computer resources
- **Protection and security** - The owners of information stored in a multiuser or networked computer system may want to control use of that information, concurrent processes should not interfere with each other
  - **Protection** involves ensuring that all access to system resources is controlled
  - **Security** of the system from outsiders requires user authentication, extends to defending external I/O devices from invalid access attempts
  - If a system is to be protected and secure, precautions must be instituted throughout it. A chain is only as strong as its weakest link.

System Calls

- ***System Calls*** *provide programming interface for user processes to request the services from the system made available by the OS.*

- **System calls** allow user-level processes to request services of the **operating system**
- Typically written in a high-level language (C or C++)
- Mostly accessed by programs via a high-level **Application Program Interface (API)** rather than direct system call use
- Three most common APIs are Win32 API for Windows, POSIX API for POSIX-based systems (including virtually all versions of UNIX, Linux, and Mac OS X), and Java API for the Java virtual machine (JVM)
- Why use APIs rather than system calls?

System Calls and API (Def.)

- *A System Call is an explicit request to the kernel made via a software interrupt by OS from user processes.*

- *Application Programming Interface (API) is a set of functions, objects, protocols or data structures for the support of application development for developers/programmers.*

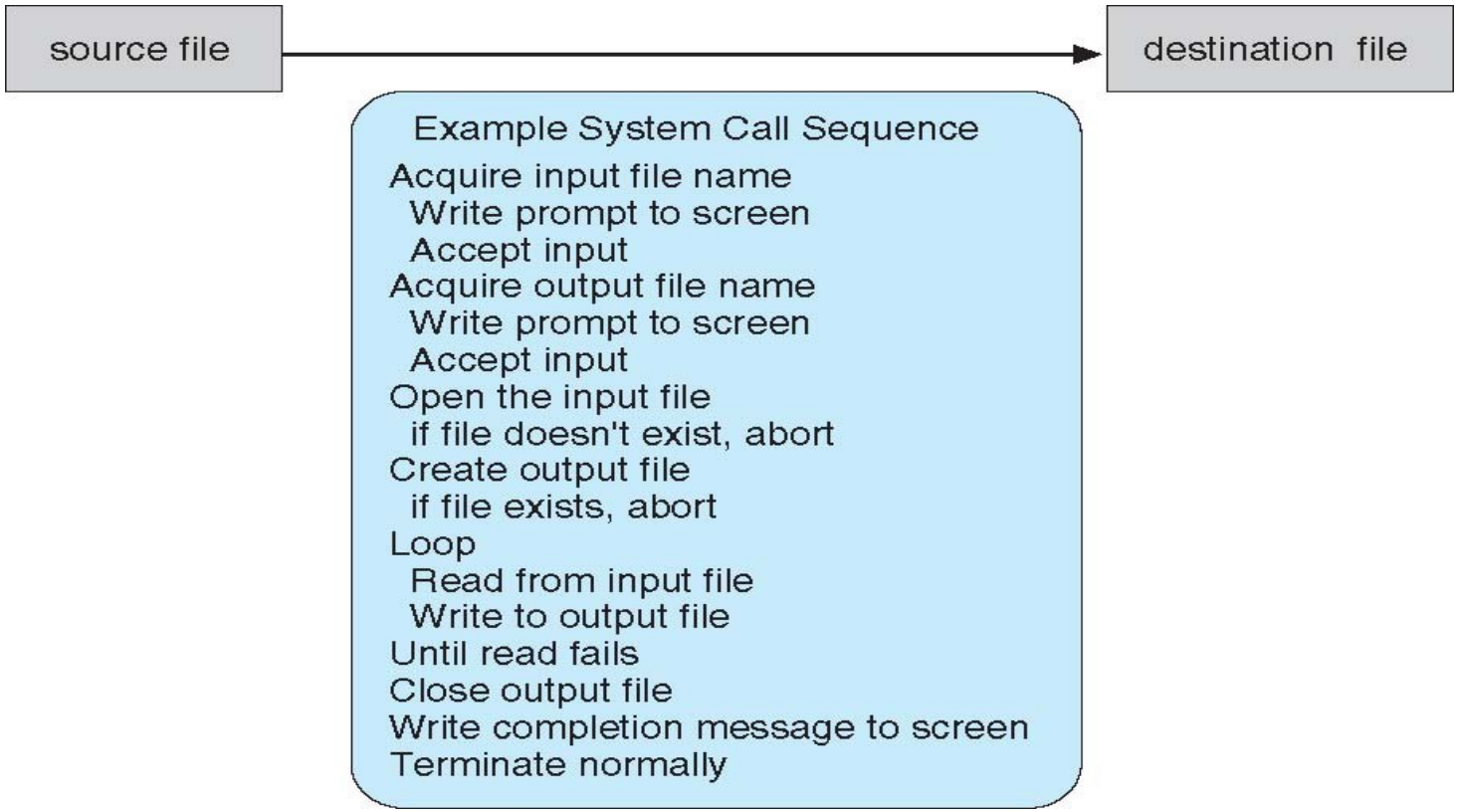
API (Def.)

- *Operating Systems provide a library (or high-level Application Program Interface or API) that sits between normal programs and the rest of the operating system.*

- *This library handles the low-level details of passing information to the kernel and switching to supervisor mode,*

Example of System Calls

- System call sequence to copy the contents of one file to another file



System Call Implementation

- Typically, a number associated with each system call
  - System-call interface maintains a table indexed according to these numbers
- The system call interface invokes intended system call in OS kernel and returns status of the system call and any return values
- The caller need know nothing about how the system call is implemented
  - Just needs to obey API and understand what OS will do as a result call
  - Most details of OS interface hidden from programmer by API
    - Managed by run-time support library (set of functions built into libraries included with compiler)

Operating System Design and Implementation

- Design and Implementation of OS not “solvable”, but some approaches have proven successful
- Internal structure of different Operating Systems can vary widely
- Start by defining goals and specifications
- Affected by choice of hardware, type of system
- *User* goals and *System* goals
  - User goals – operating system should be convenient to use, easy to learn, reliable, safe, and fast
  - System goals – operating system should be easy to design, implement, and maintain, as well as flexible, reliable, error-free, and efficient

Operating System Design and Implementation (Cont.)

- Important principle to separate

  Policy: What will be done?
  Mechanism: How to do it?

- Mechanisms determine how to do something, policies decide what will be done.
  - The separation of policy from mechanism is a very important principle, it allows maximum flexibility if policy decisions are to be changed later.

Operating System Design and Implementation

- *Selecting language for coding*
- *Assembly language was used before*
  - *Adv: faster/ run fast directly due to directly hard coding*
  - *Disadv: More no of lines with large program*
- *Now generally used C, C++ or Python.*
- *Linux and windows OS are written mainly in C*
- *Advantages of using higher-level language :*
  - *Code can be written faster*
  - *More compact*
  - *Easier to understand and debug*
  - *Improvements in compilation technology*
  - *Easier to port*

Operating System Structure

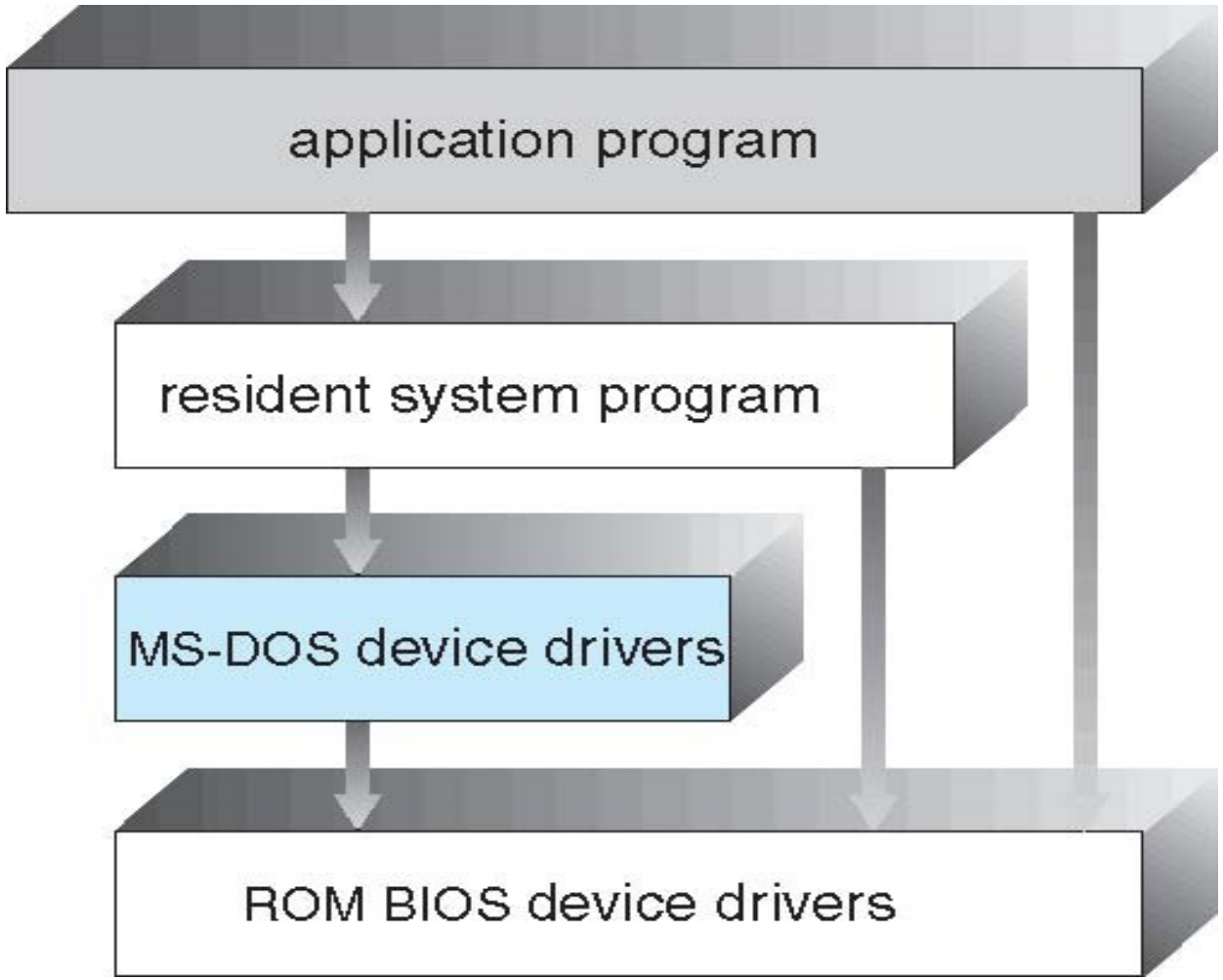
An **Operating System (OS) structure** refers to the way the operating system is organized internally. Different structures are designed to achieve goals such as high performance, modularity, security, maintainability, and portability.
There are several OS structure, such as:

1. Simple Structure
2. Monolithic Structure
3. Layered Structure
4. Microkernel Structure
5. Modular Structure
6. Hybrid Structure
7. Virtual Machine Structure
8. Exokernel Structure

1. Simple Structure

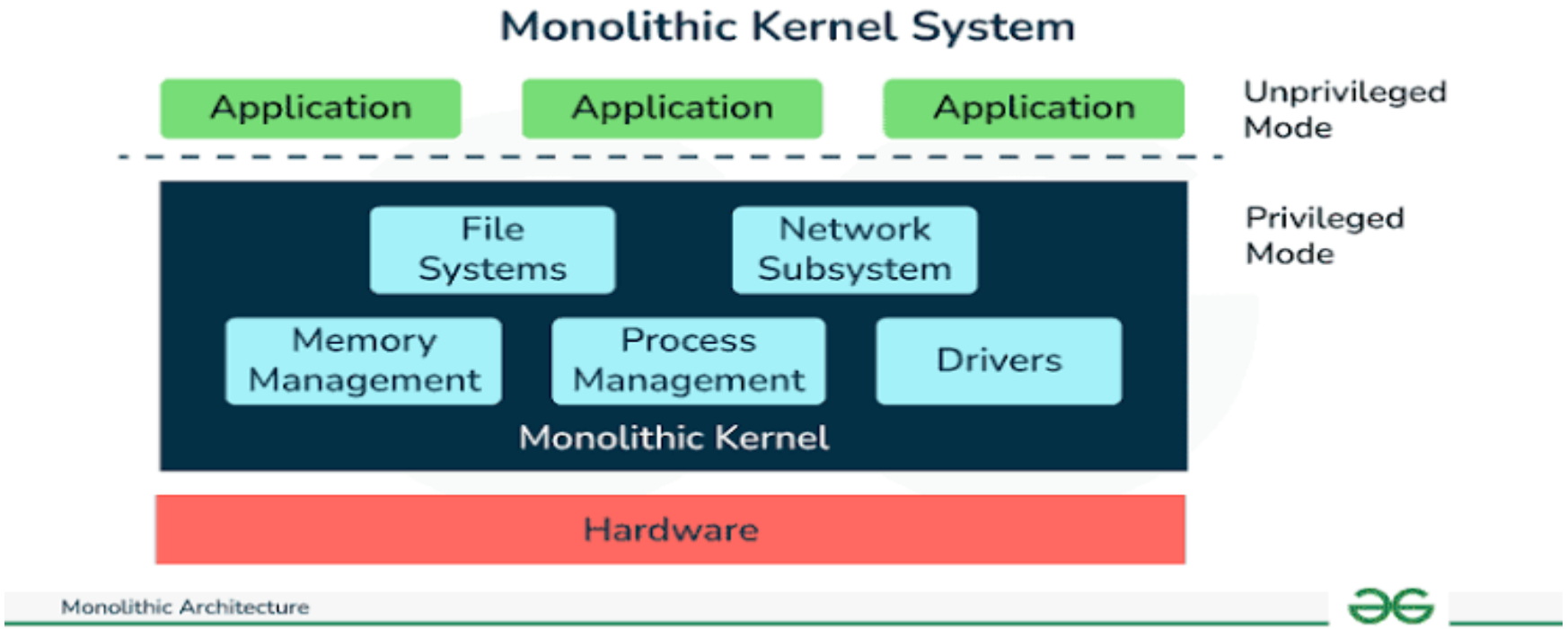
- MS-DOS – written to provide the most functionality in the least space
  - Not divided into modules
  - This kernel is hard to extend.
  - If a user wants to add a new service then the entire operating system has to be modified.
  - No dual-mode and no hardware protection
  - Leave vulnerable to errant or malicious programs
  - Crashes entire program when user program fails

MS-DOS Layer Structure



2. Monolithic Structure

In a **monolithic kernel**, all operating system services run in **kernel mode** as one large program. Every module can directly communicate with every other module.



2. Monolithic Structure

Advantages
- High performance.
- Fast communication among kernel modules.
- Efficient system call execution.

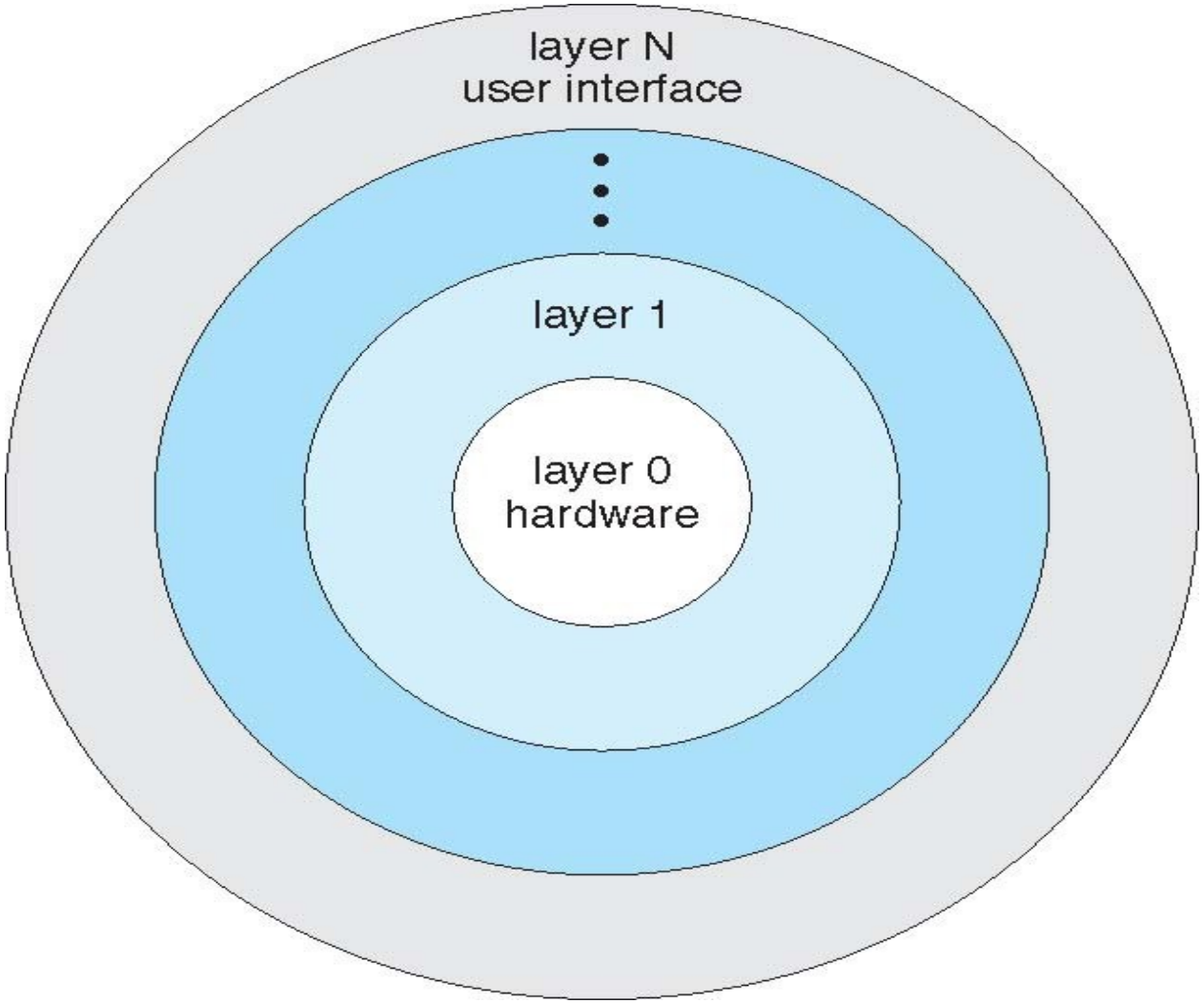
Disadvantages
- Large kernel size.
- Difficult debugging.
- A faulty driver may crash the entire OS.

Examples
- Linux
- Traditional UNIX

3. Layered Approach

- OS can be broken into smaller and more appropriate pieces creating modular operating system
- With modularity, layers are selected such that each uses functions (operations) and services of only lower-level layers
- More freedom in changing inner workings of the system

Layered Operating System



Layered Approach

- Advantages:
  - Simplicity of construction and debugging
  - Design and implementation of the system is simplified
    - Each layer is implemented only those operation provided by its lower layer
- Disadvantages:
  - Major difficulty involves appropriately defining each layer
  - Less efficient than others
    - An I/O request of user program adds overhead at each layer

4. Microkernel System Structure

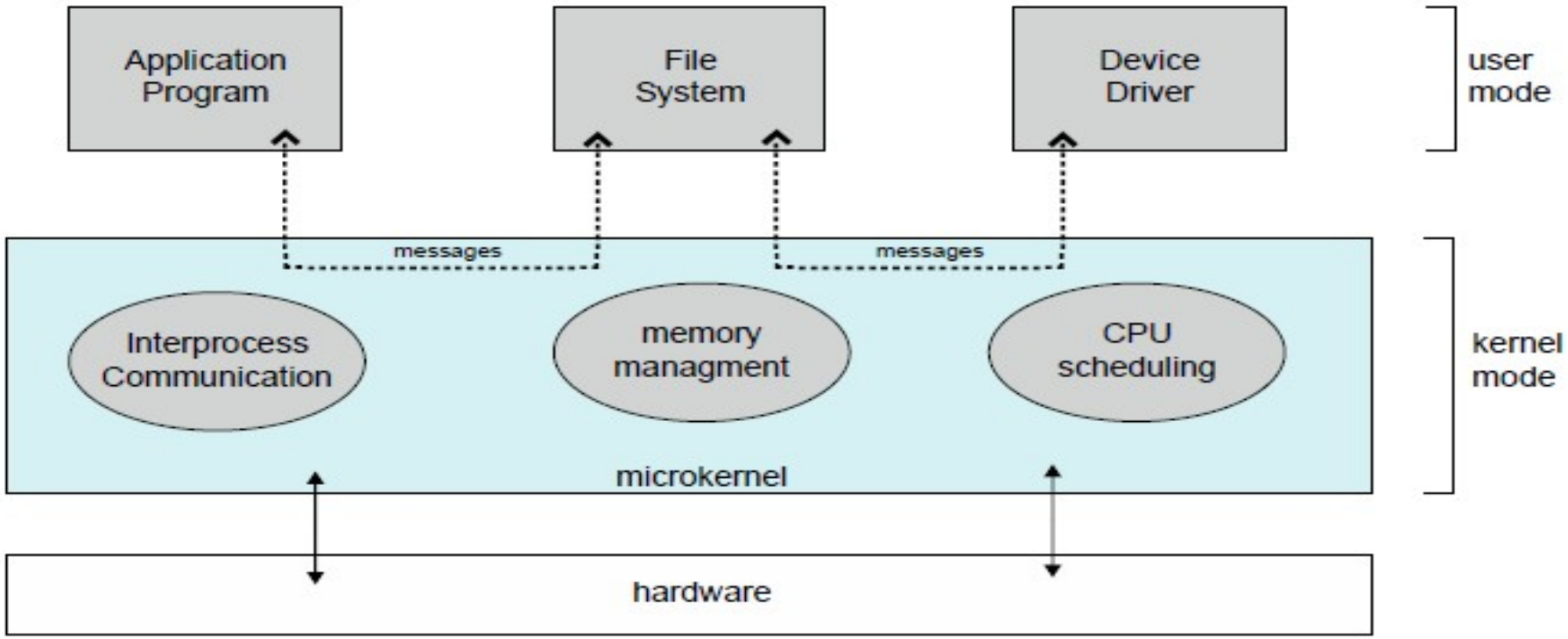
- A microkernel-based system puts only the bare minimum system components in the kernel and runs the rest of them as user mode processes.
- Moves all non-essential components as much from the kernel into "*user*" space
- Implementing as system and user level
- Communication takes place between user modules using message passing

Microkernel System Structure

- new protocol stacks, file systems, device drivers and other low-level systems located in the monolithic kernel which results to a lot of work and careful code management.

- operating system functions, such as device drivers, protocol stacks and file systems, are typically removed from the microkernel itself and are instead run in user space.
- **Examples**
  - Mach
  - Minix 3
  - QNX

Microkernel System Structure



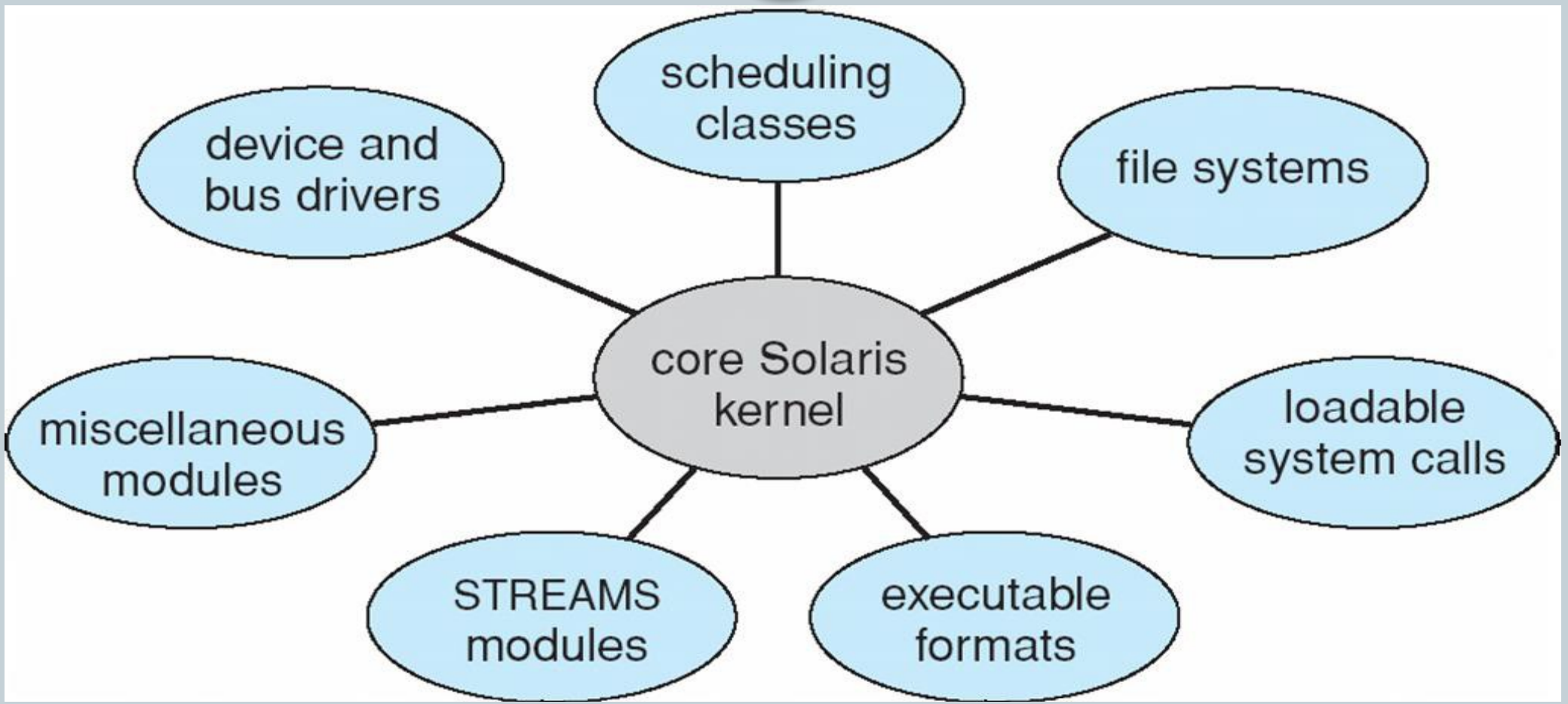
Microkernel System Structure

- Benefits:
  - Easier to extend a microkernel
  - Easier to port the operating system to new architectures
  - More reliable (less code is running in kernel mode)
  - More secure
- Demerits:
  - Performance overhead of user space to kernel space communication

5. Modular Structure

- The best current methodology for OS design is loadable kernel modules where the kernel has a set of core components and links in additional services via modules either boot time or during run time.
- Kernel is provided to design core services
- And other services dynamically is adding new features directly to the kernel (required kernel compilation every time)
- Example: Solaris, Linux and Mac OS X.

Modules Architecture



Modules Architecture

- Advantages
  - More flexible than a layered approach as one module can call another module.
  - More efficient than microkernel because modules do not need to invoke message passing to communicate
- Disadvantages
  - the fragmentation penalty is a major disadvantage of loadable modules in the kernel. This means that every time a new kernel module code is inserted, the kernel becomes fragmented.[TLB (Transaction Lookaside Buffer)]

Hybrid Architecture

- Most modern operating systems are actually not one pure model
- Hybrid combines multiple approaches to address performance, security, usability needs
- Linux and Solaris kernels in kernel address space, so monolithic(Simple), plus modular for dynamic loading of functionality
- Windows mostly monolithic, plus microkernel for different subsystem ***personalities***

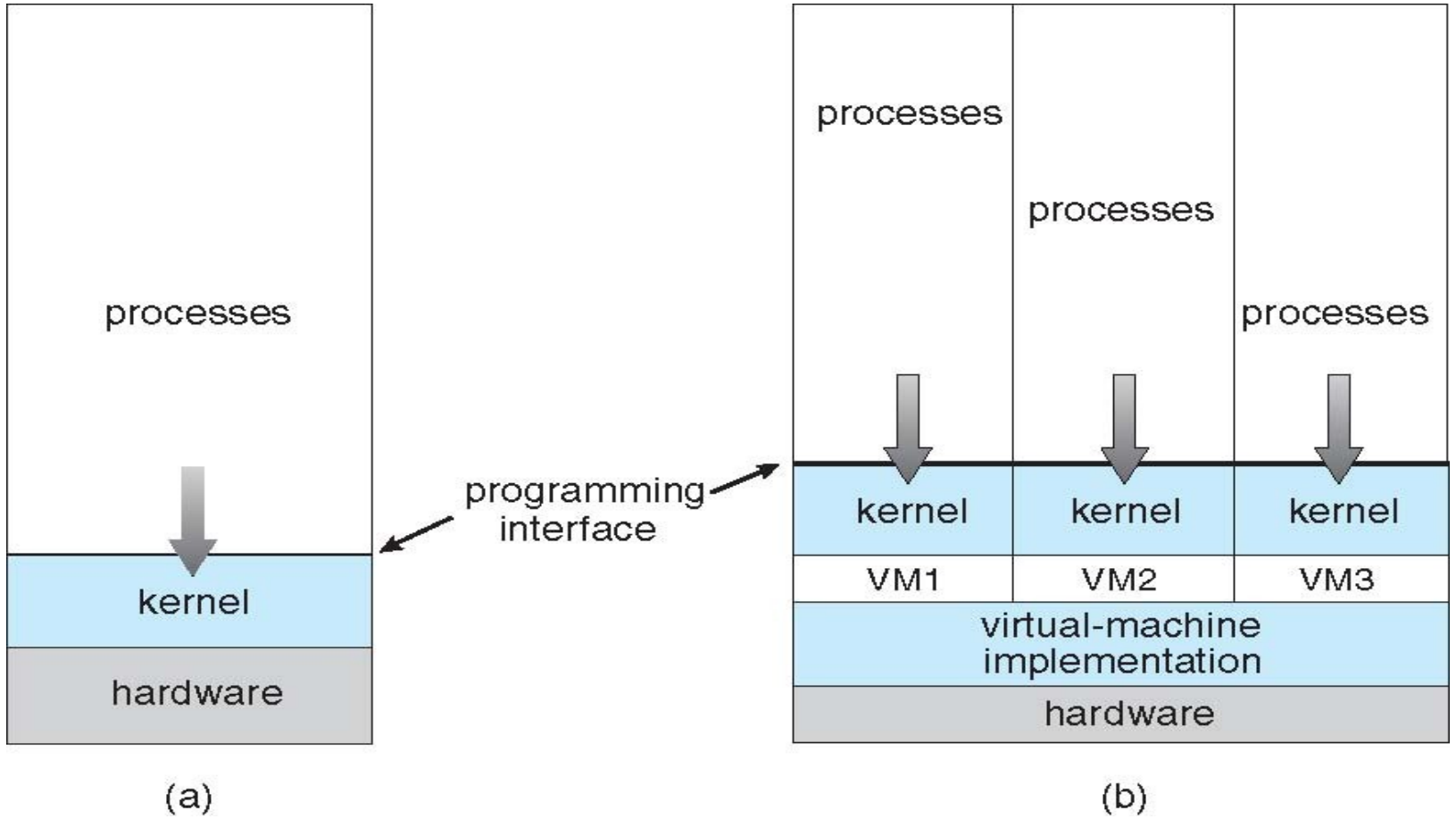
Hybrid Architecture

- Apple Mac OS X
- iOS and Android (most prominent mobile OS)

7. Virtual Machines

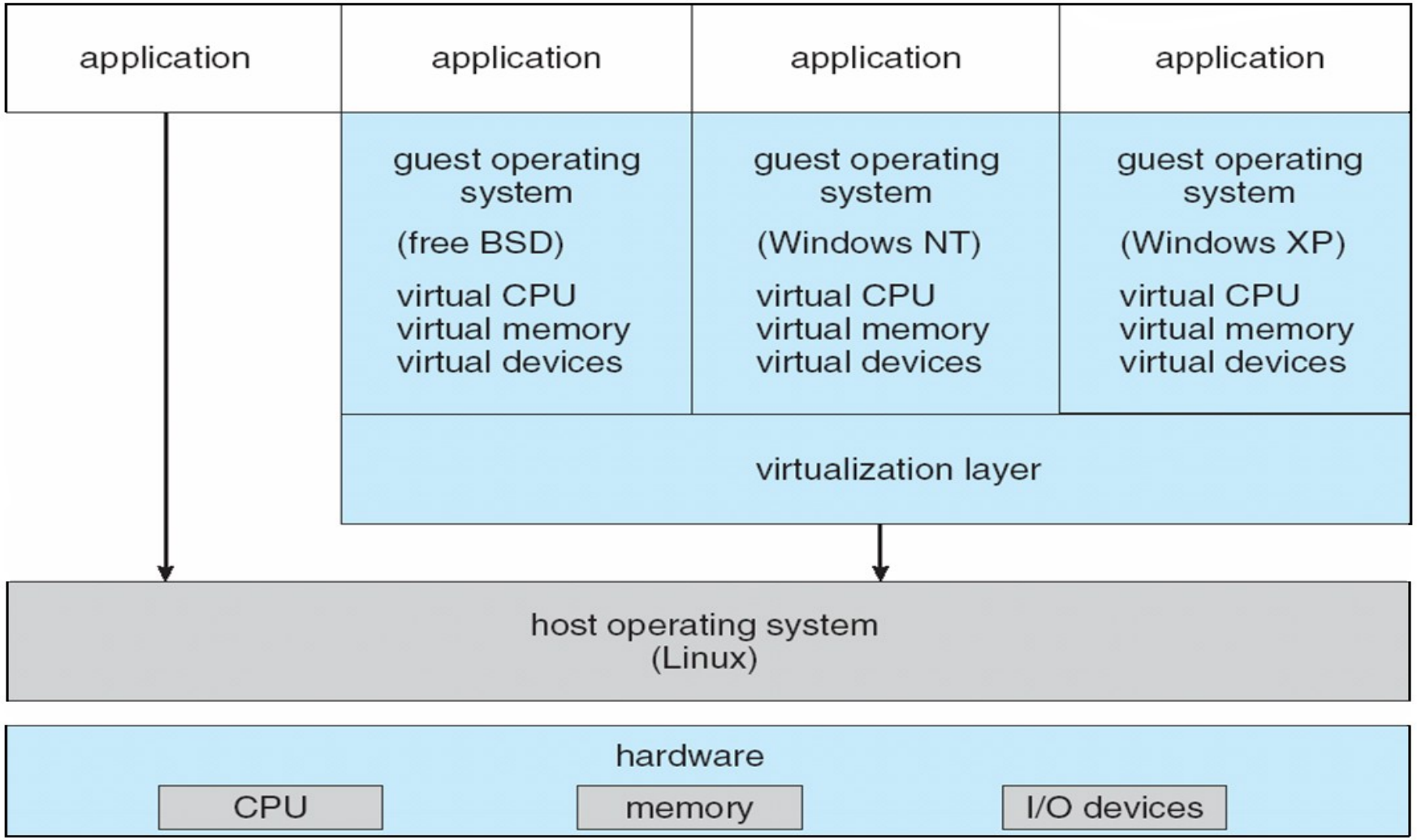
- A **virtual machine** takes the layered approach to its logical conclusion. It treats hardware and the operating system kernel as though they were all hardware.
- A virtual machine provides an interface *identical* to the underlying bare hardware.
- The operating system **host** creates the illusion that a process has its own processor and (virtual memory).
- Each **guest** is provided with a (virtual) copy of underlying computer.

Virtual Machines (Cont.)



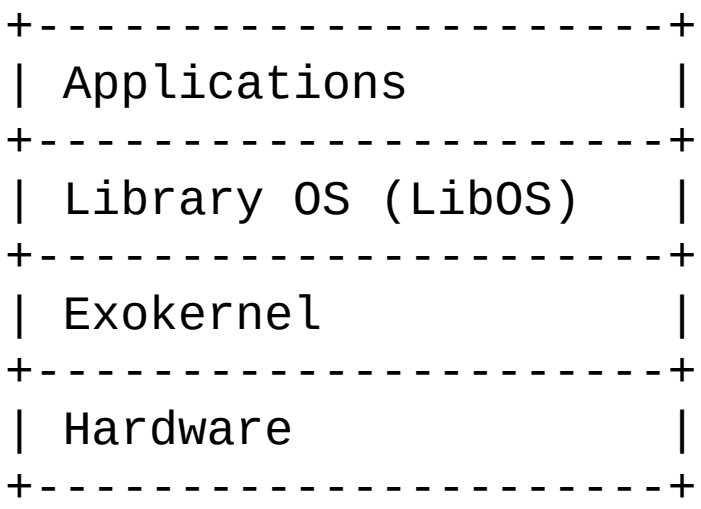
(a) Nonvirtual machine (b) virtual machine

VMware Architecture



8. Exokernel Structure

Description
The **exokernel** provides only the minimum abstraction over hardware. Applications are given direct access to hardware resources, allowing them to manage resources and make decisions that the kernel previously made. Applications manage resources through **library operating systems (LibOS)**.
Architecture



8. Exokernel Structure

Advantages
Maximum performance.
Highly customizable.
Efficient resource utilization.

Disadvantages
Difficult to program.
Rarely used in commercial systems.

Examples
MIT Exokernel
Aegis

8. Comparison Table

Structure	Performance	Security	Maintainability	Complexity	Examples
Simple	High	Low	Poor	Low	MS-DOS
Monolithic	Very High	Medium	Difficult	Medium	Linux, UNIX
Layered	Medium	High	Excellent	Medium	THE OS
Microkernel	Medium	Very High	Excellent	High	Minix, QNX
Modular	High	High	Good	Medium	Linux, Solaris
Hybrid	High	High	Good	High	Windows, macOS
Virtual Machine	Medium	Very High	Good	High	VMware, Hyper-V
Exokernel	Very High	Medium	Difficult	Very High	MIT Exokernel

End of Chapter 2